Project Initiation Report

St. George Hospital Management System

Group: G3 | Project 20 - T2 2026 | CPRO306 Capstone Project

PROJECT CHARTER

1. General Project Information	
Project Name:	St. George Hospital Management System - Centralized Mobile Healthcare Management Platform
Executive Sponsor:	Mr. Mehedi Hasan
Department Sponsor	IT Department
Impact of project	This project will allow St. George Hospital Group to replace its outdated healthcare management system with a single centralized secure mobile healthcare management platform that connects all hospital branches across different states of Australia to the headquarters, streamlining patient records, appointments, staff management and data security.

2. Project Team			
Team Members	**Student ID**	**Department**	**Email**
Manish Das	K240885	Developer (Frontend/Backend)	k240885@studen.kent.edu.au
Priya Paneru	K241081	Project Manager	k241081@student.kent.edu.au
Krishala Niroula	K240986	Tester/QA	k240986@studen.kent.edu.au
Usha Tamang	K240963	System Analyst	k240963@studen.kent.edu.au
Oyndrila Oyshi Saha	K240247	UI/UX/fronted	k240247@student.kent.edu.au

3. Stakeholders	
Primary Stakeholders	St. George Hospital Management, Staff, Patients,
Project Development Team	Manish Das, Priya Paneru, Krishala Niraula, Usha Tamang, Oyndrila Oyshi Saha
End Users:	Doctors, nurses, receptionists, daily users, patients
Mentor	Mr.Mehedi Hasan
External Service Providers	Cloud hosting provider, SMS/email gateway provider

4. Project Scope Statement

Project Purpose / Business Justification

The purpose of this project is to design and develop a centralized, secure mobile healthcare management system for St.George Hospital Group that connects all hospital branches across Australia to a single headquarter and handles their core operations. The system aims to replace the outdated system whose vulnerabilities exposed parties' personal and health records to cyberattacks.

Objectives

- Ensure data security, integrity, availability and confidentiality across all modules, with 100% role-based access control and encryption applied to all sensitive patient data.
- Move appointment scheduling online to reduce receptionist call volume, targeting at least 70% of booking made through the patient portal rather than by phone.
- Digitalize and centralize patient records to eliminate paper-based storage and cut retrieval time from a manual, multi-minute search to an instant digital search.
- Automate billing, stock and reporting workflows to reduce administrative overhead and manual data entry errors.
- Align systems design and data-handling practices with the Australian Privacy Principles (Privacy Act 1988) and the security controls referenced in the client brief (HIPAA- aligned controls, NIST framework).

- Provide real-time dashboards and reports so branch managers and admins can monitor operations and make data-driven decisions.

Deliverables

- **Improved patient registration and appointment booking process**- automated digital forms, patient account creation, and online appointment requests.
- **Internal communication and notification system**- between admin, doctors, staff and patients for appointments, results, alerts and announcements.
- **Documentation**- System Requirements Specification (SRS) covering all 14 client-specified modules and additional functional requirements, Entity Relationship Diagram (ERD) and Data Flow Diagram
- **Admin Portal** - for managing system users, branches, roles, services, inventory, and system configurations.
- Encrypted database for sensitive patient and operational data
- User manual and admin guide

Scope

This project will address:

- A four panel web application: Patient, Doctor, Hospital / Clinic and Admin panels
- Core modules: User authentication and role management, branch management, patient management, appointment and scheduling, doctor / staff management, billing basics, reporting/analytics basics
- Encryption of sensitive data in transit and at rest, and role-based access control
- Advanced AI-based analytics beyond the basic chatbot and sentiment analysis Functional Requirement 46 through Functional Requirement 48 (FR46-FR48).

This project will not include:

- Full rollout across every St. George Hospital branch - the prototype demonstrates core functionality only.
- Integration with third-party accounting or hospital finance systems.

Project Milestones	
Milestone	Week

Interim SRS Report (this report)	Week 3
Final SRS Report	Week 6
System design complete (ERD, DFD, wireframes)	Week 7
Working prototype	Week 9
Testing complete	Week 11
Final report and presentation	Week 12

Major Known Risks	
Risk	Rating
Unauthorised access to patient data	High
Data breach through weak input validation (SQL injection, XSS)	High
Exposure of payment information due to improper handlin g	High
Team members unavailable due to overlapping deadlines	Medium
Scope creep from adding features not in the original brief	Medium
Lack of consent management for data use from patients	Medium

5. Constraints

Time Constraints
The project must be completed within a limited timeframe (11 weeks), which restricts the number of the client's modules and the amount of testing that can be included in this initial version.

Infrastructure Constraints
The system will be hosted on a basic staging server environment with limited scalability. Enterprise-level deployment, mobile app integration, and advanced analytics features are outside the current scope due to infrastructure limitations.

Team and skills constraints:
The team of 5 has limited prior experience implementing AES encryption and HIPPA/NIST -aligned controls, which may extend the learning curve for the security

modules.
Testing Resource Constraints: Limited access to professional testing tools and real users means testing will be conducted manually by the development team, the mentor and select participants only, using non-real patient data.

6. System Features

The full functional and non-functional requirement list (FR1–FR48, NFR1–NFR20) will be detailed in the Final SRS Report; this interim report lists the module areas only.

Module	Module
User Authentication & Roles	Billing & Payments
Branch Management	Reports & Analytics
Patient Management & Records	Communication & Notifications
Appointment & Scheduling	Audit & Security Management
Doctor & Staff Management	Data Backup & Recovery
In-Patient (Admission & Ward)	System Configuration & Admin
Pharmacy & Inventory	Laboratory & Diagnostics

Hardware and software requirements will be recommended by the development team once system design is finalised in the Final SRS.

7. Communication Plan

In order to ensure project success, the team will communicate clearly and consistently with all stakeholders. Key goals include aligning tasks and timelines, maintaining transparency, ensuring easy access to documentation, and enabling quick sale resolution.

Methods by Stakeholders:

- Team: weekly meeting (in person or Zoom) plus daily updates on Trello and group chat.
- Lecturer: weekly progress updates and questions raised during scheduled consultation times.
- Documentation: shared through Google Drive so all members can access the latest version.

Frequency	Audience	Method	Content	Supporting Tools
Daily	Project Team Members	Whatsapp group chat, Trello	Quick task related updates and coordination	Trello board
Weekly	Project team members, Mentor	In person meeting	Progress review, planning, issue resolution	Status report added to trello board

8.Ethical Risks in Database Design and Mitigation

This system is subject to the real danger of ethical risk due to the sensitive health information that is stored. The careless construction of the database may lead to the disclosure of patient information, misuse of information, or retention for longer than required.

- Storing more personal data than needed. Mitigation: only collect fields required for care and admin, following data minimisation principles.
- Staff having wider access than their role needs. Mitigation: role-based access control so, for example, a receptionist cannot view clinical notes.
- Unencrypted storage of sensitive fields. Mitigation: encrypt data at rest and in transit using AES (National Institute of Standards and Technology, 2001).
- Keeping patient data indefinitely. Mitigation: apply a data retention policy in line with Australian health record-keeping rules.

9. Feasibility Study

- Technical: the team has the skills needed (mobile app development, database design, encryption) and the tools are freely available or already licensed through the university.
- Economic: no real budget is available since this is a student project, but all software used (development tools, hosting for testing) is free or low-cost.
- Operational: hospital staff already use appointment and record-keeping systems, so training on a new app should be manageable with a short onboarding guide.

- Legal and regulatory: the system must align with Australian privacy law, including the Privacy Act 1988 and the My Health Records Act 2012 (Australian Government, 1988, 2012), confirmed in detail during design.
- Schedule: the timeline (Section 4) is tight but achievable if the team keeps to weekly milestones and flags delays early.

10. Work Breakdown and Diagram

1. Initiation Phase

Duration: Week 1

Team: Project Manager, Analyst

- Defined project scope - a centralized mobile healthcare management system connecting multiple hospital branches to a single headquarters, replacing the client's outdated, insecure legacy system
- Identified key stakeholders across the four system panels (Admin, Doctor, Patient, Hospital/Clinic) and mapped their roles
- Established project schedule and approved project charter

Milestone: Project Kickoff Complete

2. System Design

Duration: Week 2 -3

Team: Analyst, UX Designer, Database Designer

- Consolidated functional and non-functional requirements into the SRS, including additional FRs researched and approved by the project supervisor
- Designed the ER diagram and centralized database schema, built around a global patient ID shared across all branches
- Designed a modular MVC system architecture to support future branch expansion without structural changes
- Produced UI/UX wireframes and a security design covering AES encryption, password hashing, and secure data transmission

Milestone: Design Phase Approved

3. Execution Phase (System Development)

Duration: Week 4 - 7

Team: Developers (Frontend & Backend), QA

- 3.1 Core Modules
- Authentication and role-based access control for six user types
- Patient management with global patient ID and medical history
- Appointment scheduling with conflict prevention and automated reminders

3.2 Operational Modules

- Doctor, staff and branch management
- In-patient/ward management, pharmacy and inventory tracking
- Laboratory and diagnostics with digital report handling

3.3 Administrative and AI Modules

- Billing, reporting and audit/security logging
- AI chatbot for patient queries and automated report summaries

Milestone: Core System Developed

4. Testing & Monitoring

Duration: Week 8 - 9

Team: QA, Testers, Mentor

- Unit Testing and Security Checks
- User Acceptance Testing (UAT) and Feedback Collection
- Bug Fixing and Final Adjustments

Milestone: UAT Sign-Off

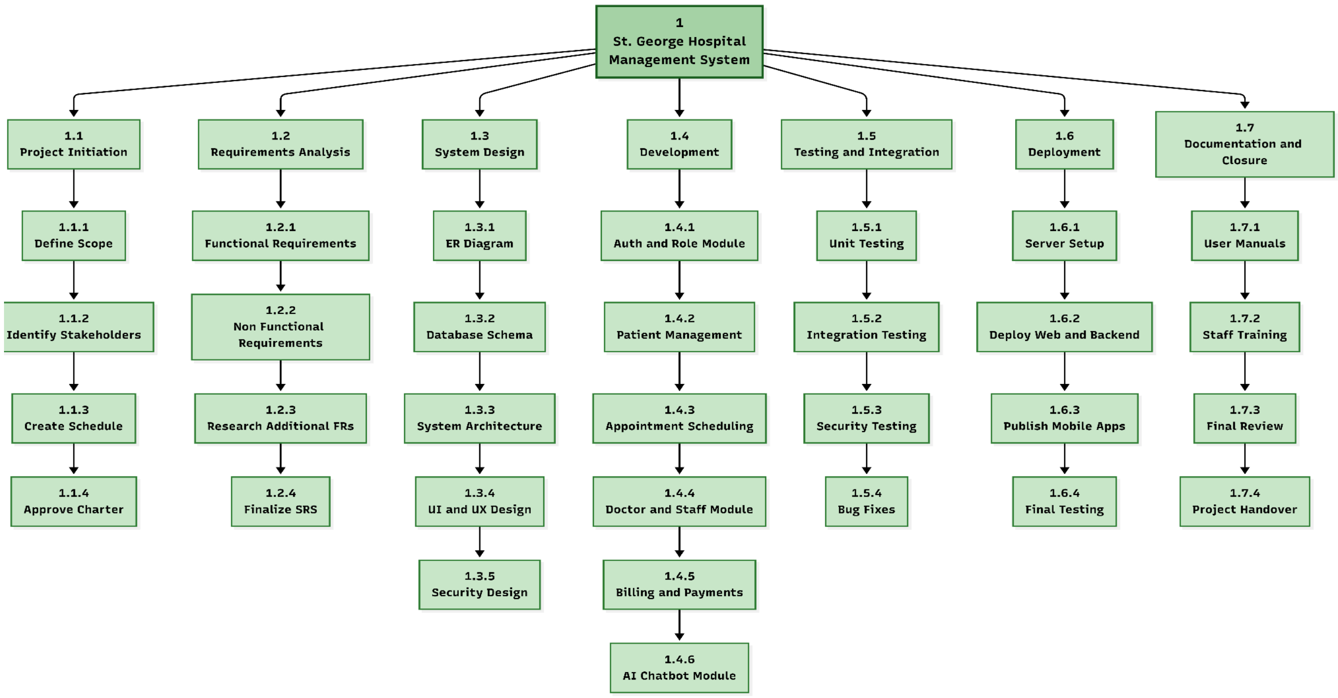
5. Deployment & Project Closure

Duration: Week 10

Team: All

- Deployment to Production
- Documentation Completion & Project Handover

Milestone: Project Completed

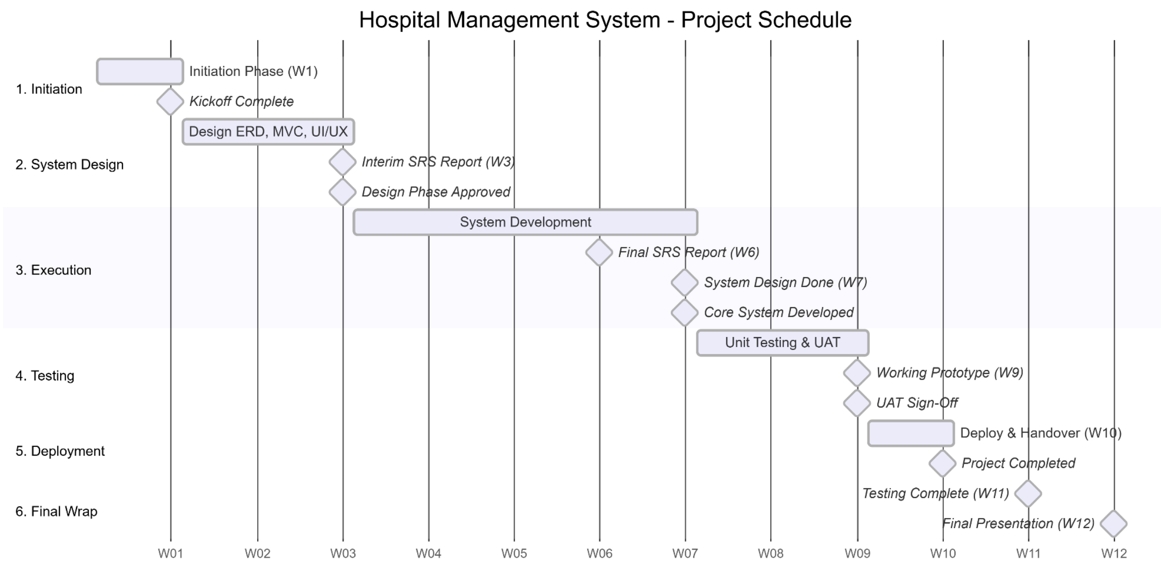


Resources

- Team members: Manish, priya, Usha, krishala, Oyndrila
- Mentor: Mr. Mehedi Hasan
- Stakeholder: Cloud hosting provider, SMS/email gateway provider

11. Gantt Chart

The Gantt chart below shows the planned timeline for our project, broken down by phase. It helped us keep track of what needed to be done at each stage and when key milestones, like the SRS approval and UAT sign-off, were expected to be completed. We used this chart to stay on schedule and make sure each phase was finished before moving on to the next one.



12. Response to Lecturer Feedback

No feedback has been received on this project yet, as this is the first formal submission. Any feedback given after this report will be addressed in the Final SRS Report submission.